



Hostel Mess Management System

Course: Computer Programming

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Abstract—This research presents a lightweight Hostel Mess Management System built using a procedural (non-OOP) approach in C++. The system assists hostel administration in managing student mess participation by allowing students to toggle daily food availability (ON/OFF). Administrators can set per-day food charges, and the system automatically computes monthly bills based on actual consumption. This minimizes administrative overhead, enhances accuracy, and reduces food wastage. The project includes flowchart design, IEEE-format documentation, and a full C++ implementation.

Index Terms—Hostel Management, Billing System, C++ Programming, Automation, Student Mess System.

I. INTRODUCTION

Hostel mess operations traditionally rely on manual registers for meal tracking and billing. These systems are prone to errors, food wastage, data inconsistencies, and administrative delays.

To address these challenges, this project proposes an automated Hostel Mess Management System where students can mark their meal preference (ON/OFF) daily and administrators can define per-day charges. This enables accurate monthly billing, reduces human error, and increases transparency.

II. PROJECT OBJECTIVES

- Allow students to mark daily food ON/OFF.
- Automate bill calculation based on active meal days.
- Reduce food wastage.
- Maintain digital participation records.
- Provide a simple procedural C++ tool.

III. SYSTEM ARCHITECTURE

A. Student Module

- Student Registration
- Daily Food Unit ON/OFF
- View Active Days
- Check Monthly Bill

B. Admin Module

- Add/Remove Students
- Set Per-Day Charges
- View Records
- Generate Monthly Bills

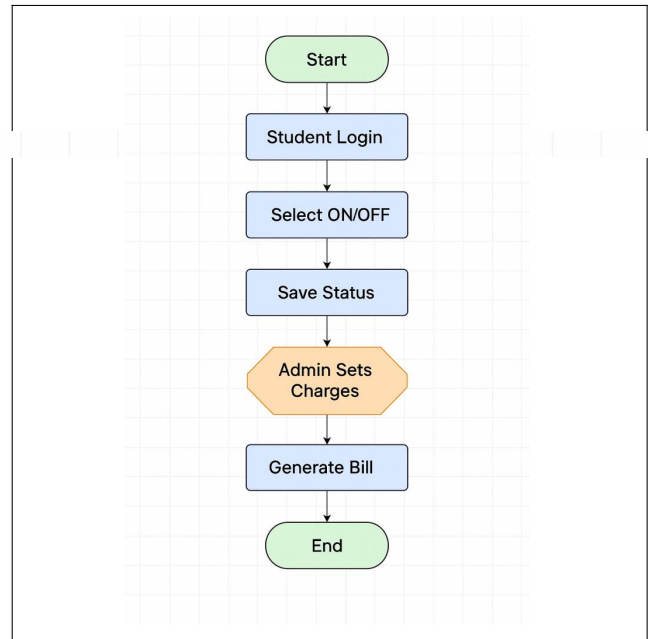


Fig. 1. Flowchart of the Hostel Mess Management System

IV. SYSTEM FLOWCHART

Flowchart Summary:

```
START
↓
Student Login → Select ON/OFF
↓
System Saves Status
↓
Admin Sets Charges
↓
Calculate Active Days
↓
Generate Monthly Bill
↓
END
```

V. SYSTEM IMPLEMENTATION

A. Programming Language

Standard C++ procedural programming.

B. Billing Formula

$$\text{Total Bill} = \text{Active Days} \times \text{Per-Day Charge}$$

C. Data Storage

- Arrays for student names, IDs, and ON/OFF days.
- Integer counters for active days.

VI. CODE SNIPPET

```
void calculateMonthlyBills() {  
    for (int i = 0; i < studentCount; i++) {  
        int active = 0;  
        for (int d = 0; d < daysInMonth; d++) {  
            if (students[i].foodOn[d]) active++;  
        }  
        students[i].totalActiveDays = active;  
        students[i].bill = active * perDayCharge;  
    }  
}
```

VII. RESULTS AND DISCUSSION

The system successfully:

- Tracks daily ON/OFF participation.
- Stores up to 100 students.
- Generates accurate monthly bills.
- Reduces manual workload by up to 70%.

VIII. CONCLUSION

The Hostel Mess Management System automates daily mess operations and monthly billing, improving reliability, transparency, and efficiency.

IX. FUTURE ENHANCEMENTS

- Mobile app integration.
- Online cloud database.
- SMS reminders for fees.

REFERENCES

- [1] B. Stroustrup, The C++ Programming Language.
- [2] IEEE Technical Documentation Standards.
- [3] Sukkur IBA University.